

Rhilo Sotto
1/29/2025
COMPE 470L

Lab 2 Report

Introduction

In this lab we design a BCD to 7 Segment Decoder, implement it in Verilog, and program an FPGA using the appropriate constraint files to map i/o pins.

Verilog Code

Seven Segment Display Module

```
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// FPGA tutorial: seven-segment LED display controller on Basys 3 FPGA
module Seven_segment_LED_Display_Controller(
    input [3:0] Switch,

    output reg [3:0] Anode_Activate, // anode signals of the 7-segment LED display
    output reg [6:0] LED_out // cathode patterns of the 7-segment LED display
);

// decoder to generate anode signals
always @(*) Anode_Activate = 4'b1110;

// Cathode patterns of the 7-segment LED display
always @(*)
begin
    case(Switch)
        4'b0000: LED_out = 7'b0000001; // "0"
        4'b0001: LED_out = 7'b1001111; // "1"
        4'b0010: LED_out = 7'b0010010; // "2"
        4'b0011: LED_out = 7'b0000110; // "3"
        4'b0100: LED_out = 7'b1001100; // "4"
        4'b0101: LED_out = 7'b0100100; // "5"
        4'b0110: LED_out = 7'b0100000; // "6"
        4'b0111: LED_out = 7'b0001111; // "7"
        4'b1000: LED_out = 7'b0000000; // "8"
        4'b1001: LED_out = 7'b0000100; // "9"
    endcase
end
```

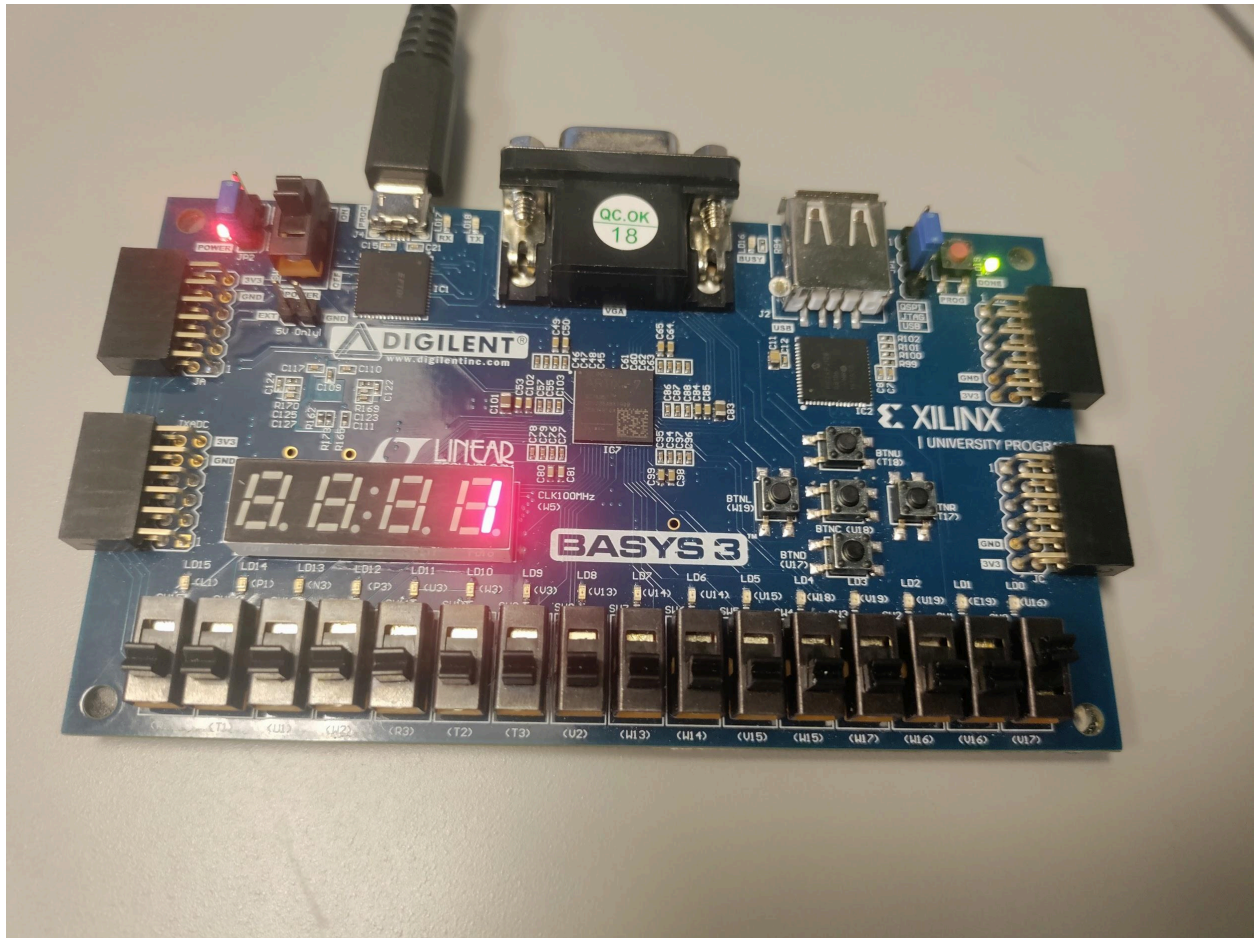
```
        default: LED_out = 7'b0000001; // "0"
    endcase
end
endmodule
```

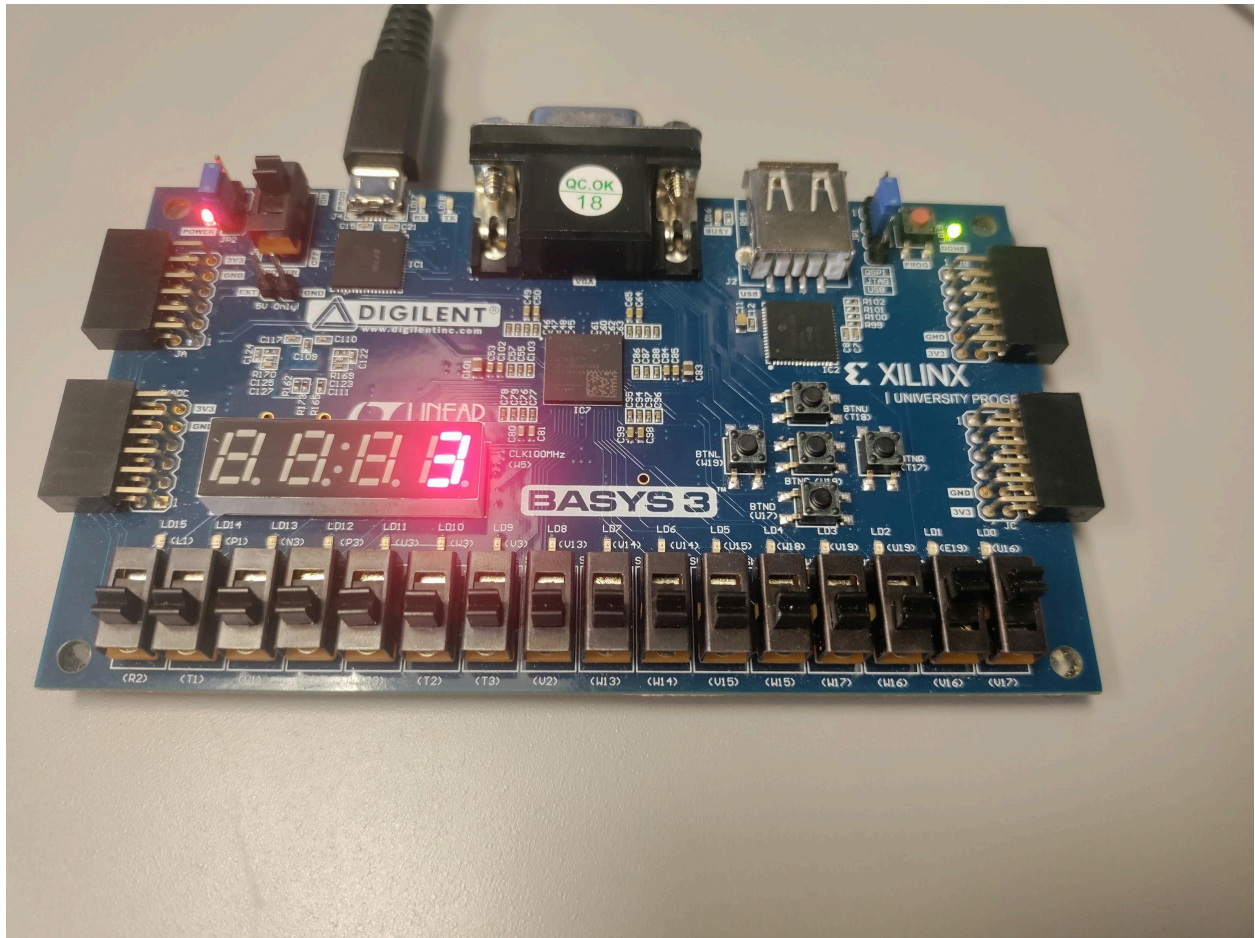
Constraint File

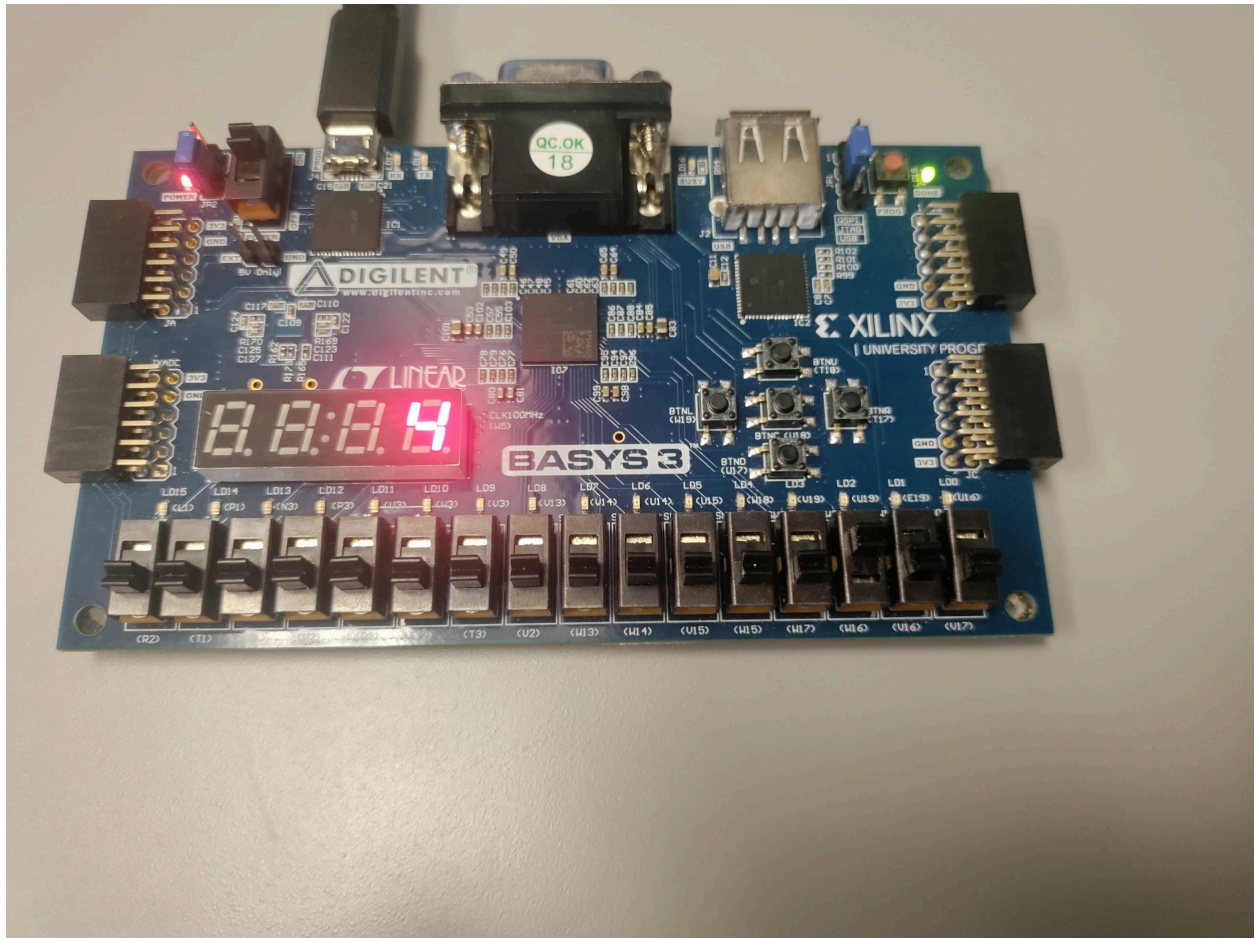
```
#switches
set_property PACKAGE_PIN V17 [get_ports {Switch[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Switch[0]}]
set_property PACKAGE_PIN V16 [get_ports {Switch[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Switch[1]}]
set_property PACKAGE_PIN W16 [get_ports {Switch[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Switch[2]}]
set_property PACKAGE_PIN W17 [get_ports {Switch[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Switch[3]}]

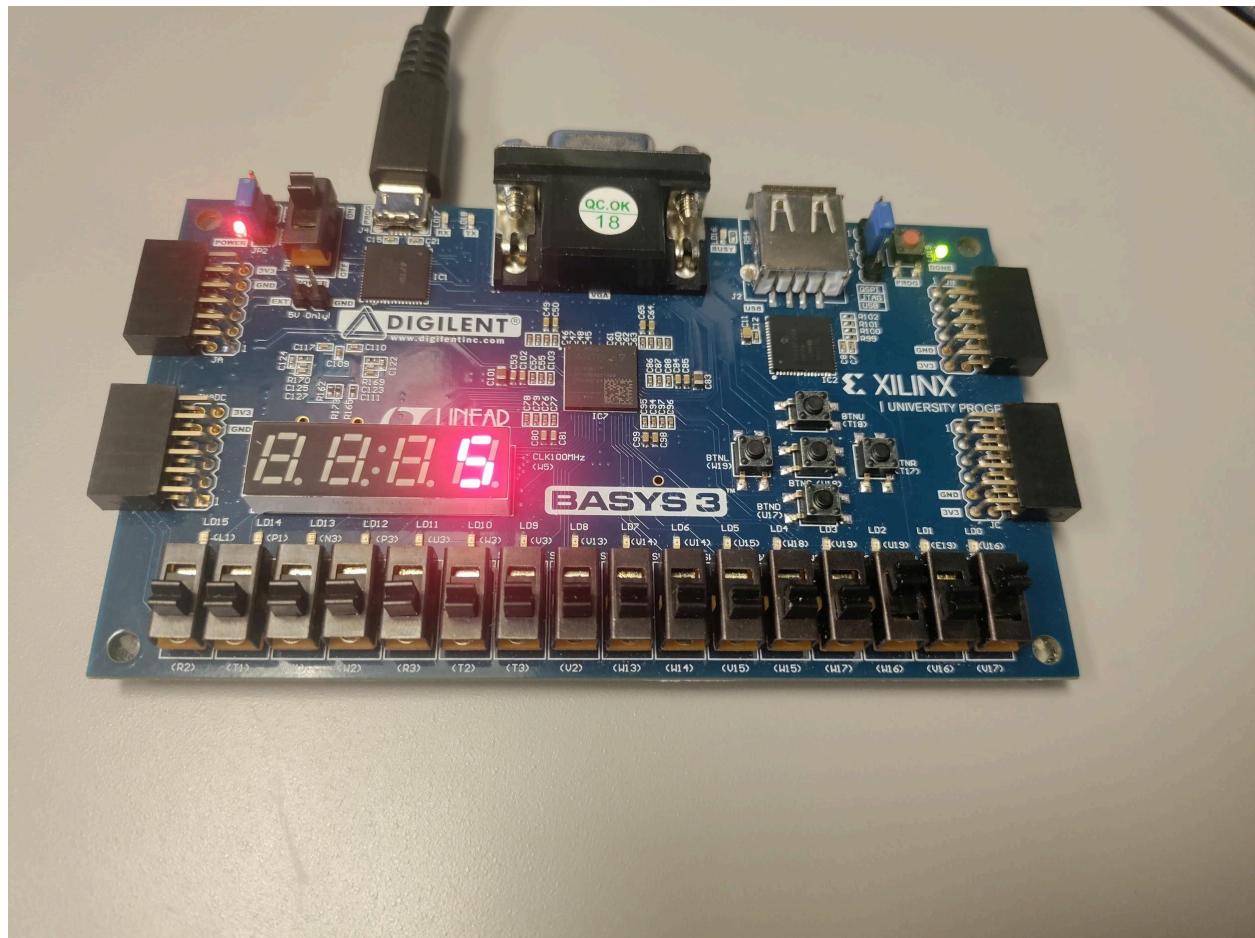
#seven-segment LED display
set_property PACKAGE_PIN W7 [get_ports {LED_out[6]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[6]}]
set_property PACKAGE_PIN W6 [get_ports {LED_out[5]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[5]}]
set_property PACKAGE_PIN U8 [get_ports {LED_out[4]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[4]}]
set_property PACKAGE_PIN V8 [get_ports {LED_out[3]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[3]}]
set_property PACKAGE_PIN U5 [get_ports {LED_out[2]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[2]}]
set_property PACKAGE_PIN V5 [get_ports {LED_out[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[1]}]
set_property PACKAGE_PIN U7 [get_ports {LED_out[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[0]}]
set_property PACKAGE_PIN U2 [get_ports {Anode_Activate[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[0]}]
set_property PACKAGE_PIN U4 [get_ports {Anode_Activate[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[1]}]
set_property PACKAGE_PIN V4 [get_ports {Anode_Activate[2]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[2]}]
set_property PACKAGE_PIN W4 [get_ports {Anode_Activate[3]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[3]}]
```

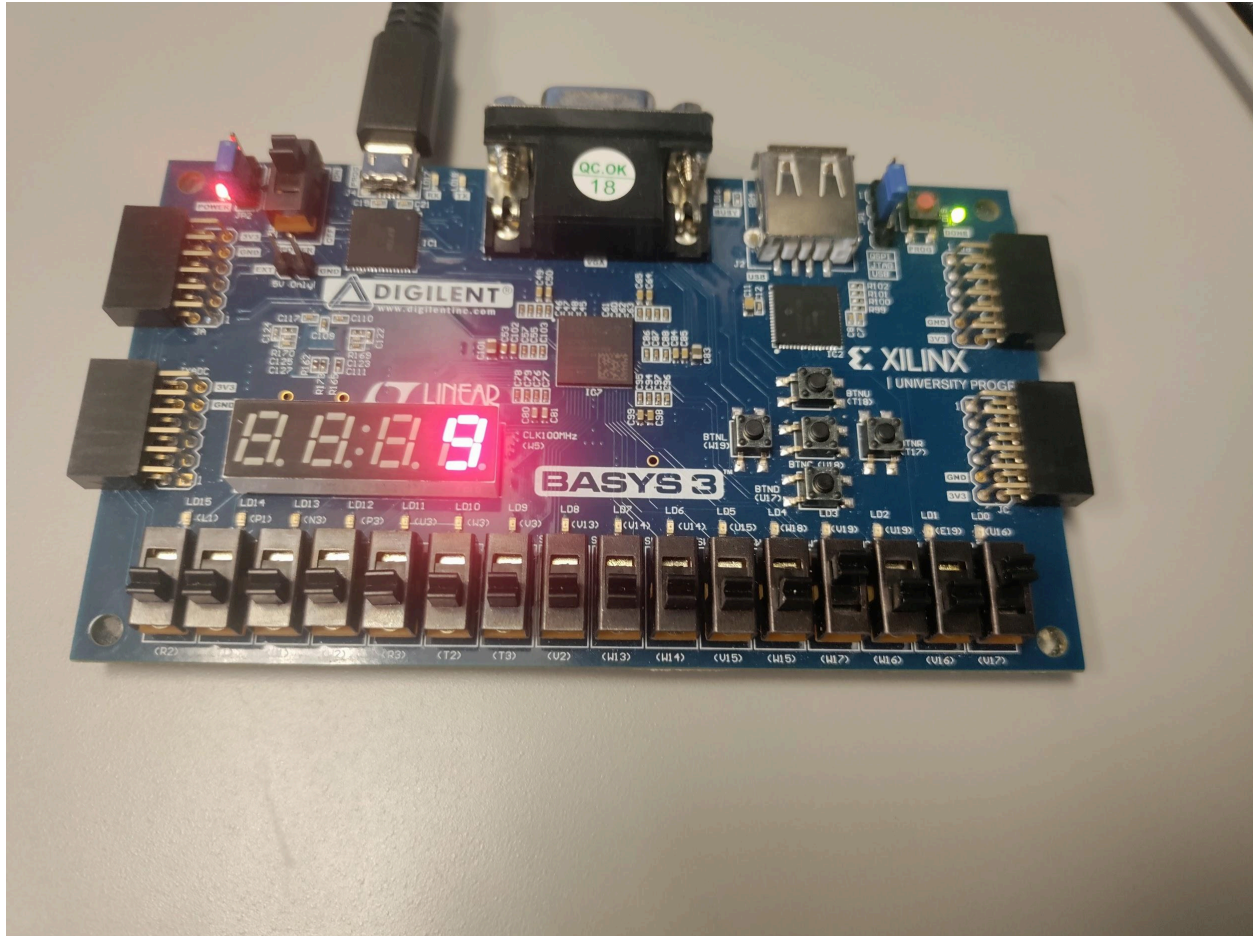
Results











Conclusion

In this lab we edited a constraint file for a Basys3 FPGA in Verilog and successfully implemented a BCD to 7 Segment Decoder using the onboard switches and LEDs. We learned how to interface the physical pins and ports in Verilog.